

Q1

- a. Vectorbase / **Vector Database** / Vectorstore.
- b. Embedded dense vector / **Embeddings** (also raw text, metadata) .
- c. **Similarity search** (nearest neighbor, cosine, dot-product or euclidean distance ...)

Award points if the answers contain the above, even if in slightly different terms.

Q2

- a. (1pt) mention self (**AUTONOMOUS**) decision-making and planning.
The answer needs to contain the concept of autonomy/independence/"agency".
- b. (1pt) **Memory** (space to store the database & information) ['workspace']
(1pt) **Tools**/Hardwares/Sensors (to access the external information) ['backend']
The answer DOES have to contain the idea of agents using local memory/cache/context etc and MUST mention the use of tools

Q3

(1 pts.) a. Model Context Protocol (**MCP**)

(1 pts.) Abstracts technical complexity standardized for agents to use "tools", ie. files/DBs/APIs/search/services/... etc.

(1 pt.) b. Musketeer - an **intermediate** representation [similar to MCP which is an intermediate 'routing layer']. Answer MUST mention Musketeer [not something more general such as dataflow or DAG or graph etc]

Q4

- 1. **Lack of true transformation**(1 pts) – The use of copyrighted data in AI training is **not meaningfully transformative**; many AI outputs **replicate the style, structure, or substance** of the original works (e.g., mimicking **Taylor Swift's** songs).
- 2. **Commercial use and market harm** (1 pts)– AI companies (like Meta, OpenAI, Midjourney) use the data for **commercial gain**, and the outputs can **substitute or**

compete with original works, causing **market harm**, which weighs **against fair use**.

3. **Large-scale, unlicensed copying**(1 pts) – These models rely on **massive, systematic copying** of **creative, copyrighted material** (like news, images, music) without permission, violating the “**amount and substantiality**” and “**nature of the work**” criteria in fair use analysis.

Point #2 above [COMMERCIAL GAIN] is the main sticking point, so if an answer discusses just that, it still gets full points.

Q5: 3 pts

Sample Answer

These days, graph databases like Neo4j are to enhance Large Language Models (LLMs) through Retrieval-Augmented Generation (RAG) architectures and agentic AI systems. When a user asks a question, the Neo4j systems retrieve not only similar text but also traverses the graph to find the relevant information and their connections. They enable LLMs to generate more accurate and trustworthy outputs in their interconnected knowledge graphs (workspace) and database operations (backend) beyond simple database lookups.

TL;DR: graphs now serve the purpose of being **knowledge graphs in RAG**, ie. in “graph RAG”.

An answer that discusses KGs and their use in RAG, is on the right track.

System Architecture: LLM + Neo4j Knowledge Graph

...

User Prompt



[Orchestrator / Agentic Layer]



[Neo4j Knowledge Graph] ↔ [Vector Search Index]



[LLM (e.g., GPT, Gemini)]



Generated Response (grounded in graph data)

...

Score Distribution

- 3 pts: Clearly explains how Neo4j or a graph database fits into RAG and Agentic AI systems.
- 2 pts: Only states that Neo4j or a graph database is used with LLMs, without explaining how or why. Or lacks of explaining RAG and Agentic AI.
- 1 pts: Mentions some of the key words such as Neo4j, a graph database, LLMs, RAG, or Agentic AI but shows no connection among them. Or shows a significant lack of understanding of those concepts. Or explains incorrectly.
- 0 pts: No writing

Q6

General Requirements:

- The JSON must be valid (proper braces, brackets, commas, colons).
- The data must correctly reflect what is shown in the question images.
- Minor format variations are acceptable if meaning and correctness are preserved.

Example:

A. Semantic triplet

```
{  
  "John": {  
    "born-in": "1945"  
  }  
}
```

OR

```
{  
  "subject": "John",  
  "predicate": "born-in",  
  "object": "1945"
```

```
}
```

B. A timer IC

```
{  
  "timerIC": {  
    "model": "555",  
    "pins": {  
      "1": "Ground",  
      "2": "Trigger",  
      "3": "Output",  
      "4": "Reset",  
      "5": "Control Voltage",  
      "6": "Threshold",  
      "7": "Discharge",  
      "8": "VCC"  
    }  
  }  
}
```

OR

```
{  
  "timerIC": {  
    "1": "Ground",
```

```
"2": "Trigger",  
"3": "Output",  
"4": "Reset",  
"5": "Control Voltage",  
"6": "Threshold",  
"7": "Discharge",  
"8": "VCC"  
}  
}
```

Or "555" as the overall key [instead of "timerIC" above]

C. Pixel

```
{  
  "pixel": {  
    "x": 4,  
    "y": 2,  
    "red": 241,  
    "green": 252,  
    "blue": 23  
  }  
}
```

OR

```

{
  "pixel": {
    "position": {
      "x": 4,
      "y": 2
    },
    "color": {
      "red": 241,
      "green": 252,
      "blue": 23
    }
  }
}

```

The answer MUST contain r,g,b AND x,y data.

Q7

A

- Clearly explains that **B-tree or similar hierarchical index structures** organize data in a balanced tree structure that allows for binary search-like operations, reducing the search space by half at each step. (1 pt)
- Presents a clear, labeled diagram showing a B-tree or similar structure with multiple levels. (1 pt) *[Note from Nima: the diagram should also show halving at each level]*

B

- Correctly identifies and explains that **hash-based** indexes can achieve $O(1)$ time complexity. (1 pt)

Q8

- As the reference to BI is given in the question, a correct brief explanation of BI with the data-driven insights, and also comparing the data usage for both AI and BI. **(1.0 point)**
- Clearly explains how supervised ML uses labeled data to learn patterns and predict future outcomes. Must mention **labels**, **training (via backprop)**, and **prediction**. **(1.0**

point)

- Partial or not completely satisfying the process **(0.5 point)**

- The diagram that shows how supervised ML works (data → model → prediction). Must be relevant and labeled. **(1 point)**

Sample Answer:

- Business Intelligence (BI) marked the beginning of using large amounts of data to generate data-driven insights. It focuses on analyzing historical data to identify trends, patterns, and summaries, helping organizations understand what happened in the past through reports, dashboards, and visualizations.
- In contrast, Artificial Intelligence (AI) — particularly supervised machine learning — also depends heavily on data but uses it differently. Supervised ML works by training a model using a dataset where each input example (called a feature) is paired with a known correct output (called a label). The algorithm learns a mapping from inputs to outputs — this is known as the training process.
- Once the model is trained, it can be used to make predictions on new, unseen data by applying the patterns it has learned. This process allows the system to behave intelligently by making data-driven decisions. Thus, while BI uses data to describe the past, supervised ML uses labeled data to learn and predict future outcomes.

Supervised Machine Learning (AI)

Input: Labeled Data (X: Features, Y: Labels)



Train ML Model



Predict Y for New X (Unseen Data)



"Predict what will happen next."

(Predictive Analytics)

Q9 Accurate Explanation of Both Indexing Types. - 1pt

Diagram - 1pt

Clear Connection Between the Two - 1pt

- 1 mark – Spatial Indexing: Correctly explains spatial indexing (e.g., R-tree) with a simple 2D diagram showing spatial regions and how queries are optimized.
- 1 mark – LLM-related Indexing: Correctly explains vector indexing (e.g., FAISS/ANN) with a diagram showing **vectors in high-dimensional space** and similarity search.
- 1 mark – Connection: Clearly relates both methods as ways to index and search based on **proximity** (geometric or semantic), showing conceptual similarity.

TL;DR: the question is about how both these indices (spatial data, vector embeddings) related to **SPATIAL/DISTANCE similarity** (not bit similarity, set elements similarity etc etc). An answer that discusses this is on the right track.

Q10

(+3 points) mentioning SQL cache/procedure **cache** and correct explanation of the working.

The answer can include the following points -

1. The SQL cache stores the most recently used SQL statements or PL/SQL procedures
2. DBMS retrieves data from the permanent storage and stores in the RAM
3. This is efficient since working with data cache is faster than working with data files

(+2 points) mentioning the concept/method but incorrect/partial explanation

(+0 points) mentioning incorrect method/incorrect explanation

In other words, the correct answer would discuss **SQL caches in the context of performance tuning for relational databases.**